

WASP-12b

R-1795

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160213 · created 2026-08-03T01:18:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457431.79955 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.044
Transit depth (Rp/Rs)^2	2.37 +/- 1.37 [%]
Orbital Inclination (inc)	79.91 +/- 2.84
Airmass coefficient 1 (a1)	1.03 +/- 0.61
Airmass coefficient 2 (a2)	-0.01 +/- 0.055
Scatter in the residuals of the lightcurve fit is	56.96 %
Best Comparison Star	None
Optimal Aperture	3.17
Optimal Annulus	7.56
Transit Duration (day)	0.12 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.044 vs 0.1178 (deviation 0.72σ)
Duration fit vs published	0.12 d vs 0.125 d (deviation 0.44σ)
Mid-time O–C	-0.1 min
Depth SNR	3.1
Residual scatter	56.96 %

Light curve

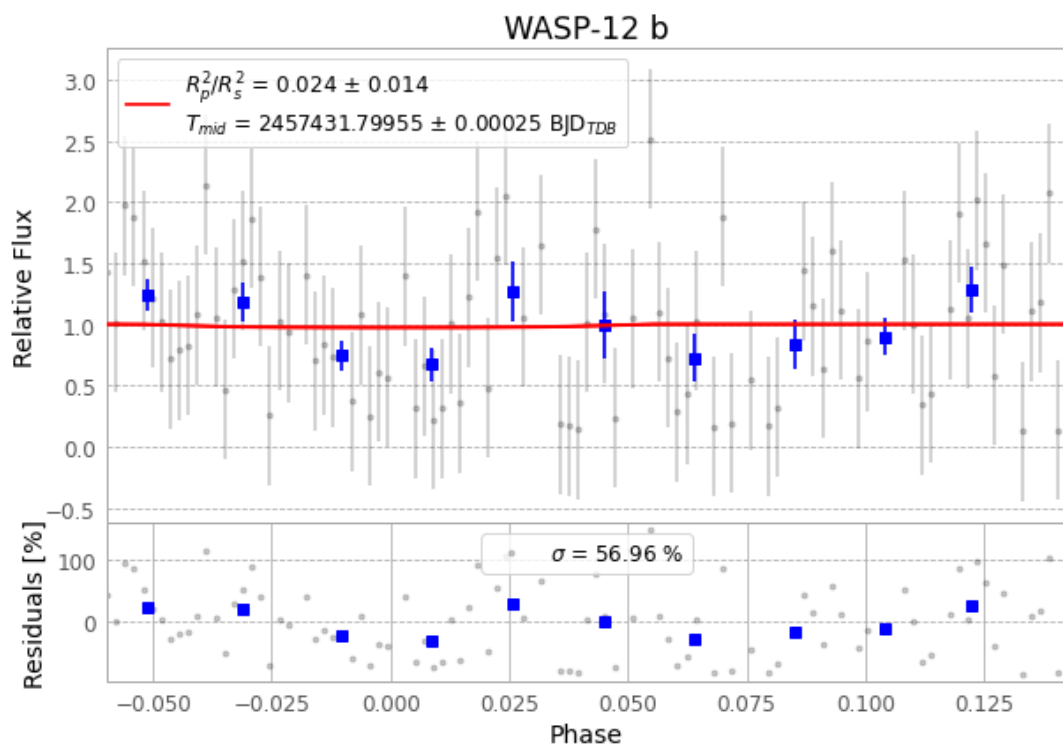


Figure 1 — FinalLightCurve_WASP-12 b_2016-02-12.png (SHA-256 88d593512e223ead...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [486, 371], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e5e51fee4b675349...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), WASP-12160213053012.FITS → WASP-12160213104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160213.log (c80cc439438b...), FinalLightCurve_WASP-12 b_2016-02-12.png (88d593512e22...), FinalLightCurve_WASP-12 b_2016-02-12.csv (ea2855df791f...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 01:18Z.